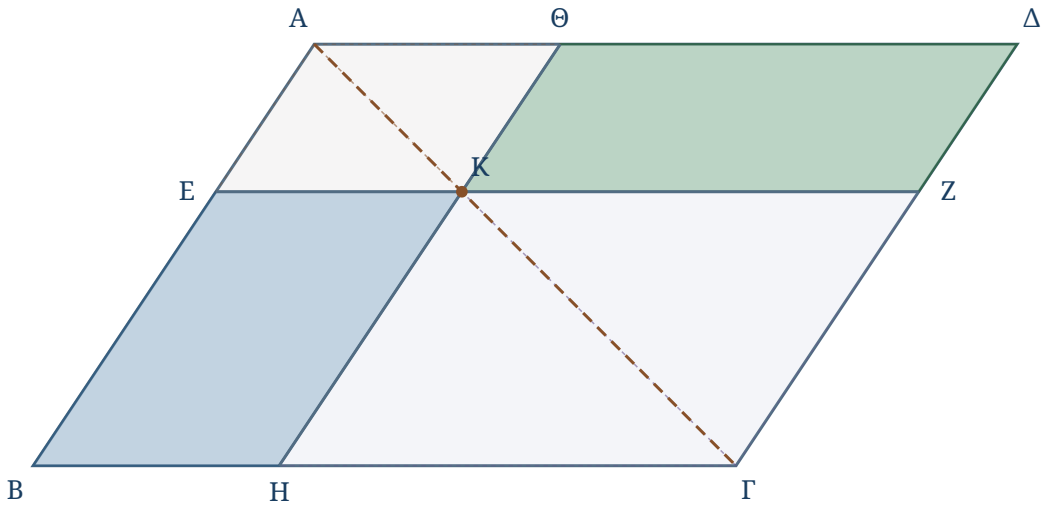
Equal leftovers. Unequal shapes.

A saved experiment with both complete proofs and ten printed source pages.

Open these settings in the workshop · Save the measurements CSV

I.43: what the diagonal makes possible

Cut across: 35%. Cut down: 35%. Upper-side shift: 4 units. K lies on AΓ; the complements have equal area.



K lies on the diagonal AΓ

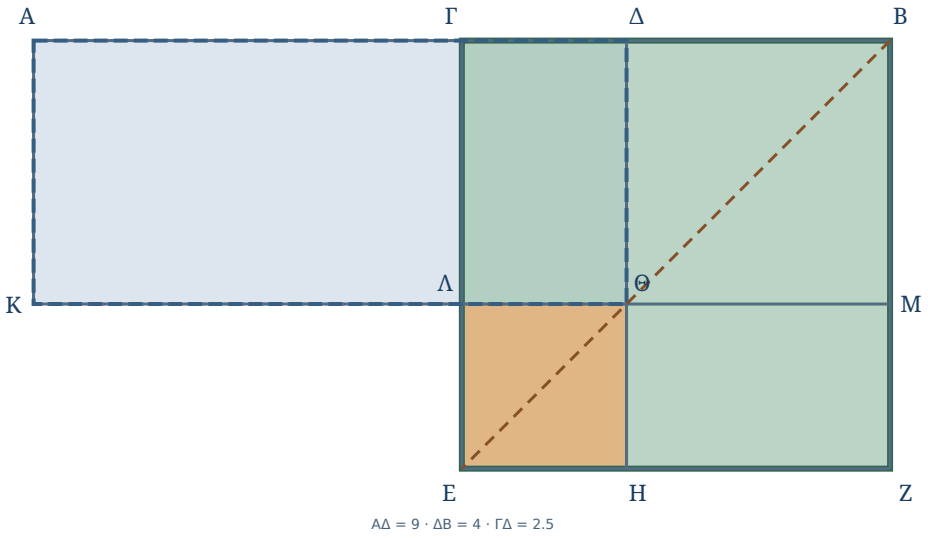
Areas in square drawing units

Region	Area	Reason
Whole triangle ABΓ	30	Half the parallelogram, I.34
Whole triangle AΔΓ	30	The other half
AEK + KHΓ	16.35	One small triangle from each pair
BK	13.65	30 − 16.35
KΔ	13.65	30 − 16.35

The large diagonal halves have equal area by I.34. When K is on that diagonal, subtracting the two pairs of equal small triangles leaves equal complements. When K is off it, the small pairs still have equal areas, but they no longer partition the large halves as required. The drawing preserves your selected stage; the full account remains available below.

II.5: the missing square

Unequal lengths AΔ = 9, ΔB = 4. Rectangle and gnomon area 36; missing square area 6.25; whole square area 42.25.



Follow the equal additions in II.5

Step	Left account	Right account
Equal complements	ΓΘ: 10	ΘZ: 10
Add ΔM	ΓM: 26	ΔZ: 26
Replace ΓM with AΛ	AΛ: 26	ΔZ: 26
Add ΓΘ	AΘ: 36	Gnomon MNΞ: 36
Add ΛH	36 + 6.25	Whole square: 42.25

The diagram follows the source's point names and orientation. The highlighted areas are compared in place; overlapping highlights name a common region, not overlapping physical paper. The gnomon MNΞ is the union ΓΘ + ΔZ. It is not a polygon with just those three vertices.

Modern coordinates: A is (0, 0), downward is positive y

Point	x	y
A	0	0
Γ	6.5	0
Δ	9	0
B	13	0
K	0	4
Λ	6.5	4
Θ	9	4
M	13	4
E	6.5	6.5
H	9	6.5
Z	13	6.5

Complete I.43: Greek, Latin and English

1. The leftovers are equal

Παντὸς παραλληλογράμμου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν.

In quouis parallelogrammo complementa parallelogrammorum circum diametrum positorum inter se aequalia sunt.

In every parallelogram, the complements of the parallelograms about the diagonal are equal to one another.

παραπληρώματα / complementa names the two remaining regions. ‘Equal’ compares their areas. Their shapes and side lengths need not match. διάμετρος here is a diagonal of a parallelogram, not the diameter of a circle.

Printed pages 100, 101.

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2. Which two regions?

Ἔστω παραλληλόγραμμον τὸ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ ἡ ΑΓ, περὶ δὲ τὴν ΑΓ παραλληλόγραμμα μὲν ἔστω τὰ ΕΘ, ΖΗ, τὰ δὲ λεγόμενα παραπληρώματα τὰ ΒΚ, ΚΔ· λέγω, ὅτι ἴσον ἐστὶ τὸ ΒΚ παραπλήρωμα τῷ ΚΔ παραπληρώματι.

Sit parallelogrammum ΑΒΓΔ, diametrus autem eius ΑΓ, et circum ΑΓ parallelogramma sint ΕΘ, ΖΗ, et complementa, quae uocantur, ΒΚ, ΚΔ. dico, esse ΒΚ = ΚΔ.

Let ΑΒΓΔ be a parallelogram, with diagonal ΑΓ. Let ΕΘ and ΖΗ be the parallelograms about ΑΓ, and ΒΚ and ΚΔ the regions called complements. I say that the complement ΒΚ equals the complement ΚΔ.

Two opposite corners can name an entire parallelogram: ΒΚ is an area here, not just the segment between Β and Κ. The construction places Κ on ΑΓ. Freeing Κ from that diagonal changes the hypothesis, even if the four regions still look like parallelograms.

Printed pages 100, 101.

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3. The diagonal divides the whole

Ἐπεὶ γὰρ παραλληλόγραμμόν ἐστι τὸ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ ἡ ΑΓ, ἴσον ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΓΔ τριγώνῳ.

nam quoniam parallelogrammum est ΑΒΓΔ, et diametrus eius ΑΓ, erit △ ΑΒΓ = ΑΓΔ.

Because ΑΒΓΔ is a parallelogram and ΑΓ its diagonal, triangle ΑΒΓ equals triangle ΑΓΔ.

I.34 supplies the equal halves. The model can shear the parallelogram while keeping its base and height fixed. That is a modern way to inspect the generality of the argument; the proof does not depend on measuring these triangles.

Printed pages 100, 101.

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4. Two smaller pairs

πάλιν, ἐπεὶ παραλληλόγραμμόν ἐστι τὸ ΕΘ, διάμετρος δὲ αὐτοῦ ἐστιν ἡ ΑΚ, ἴσον ἐστὶ τὸ ΑΕΚ τρίγωνον τῷ ΑΘΚ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΚΖΓ τρίγωνον τῷ ΚΗΓ ἐστιν ἴσον.

rursus quoniam parallelogrammum est ΕΘ, et diametrus eius ΑΚ, erit △ ΑΕΚ = ΑΘΚ. eadem de causa etiam ΚΖΓ = ΚΗΓ.

Again, ΕΘ is a parallelogram with diagonal ΑΚ, so triangle ΑΕΚ equals triangle ΑΘΚ. For the same reasons, triangle ΚΖΓ also equals triangle ΚΗΓ.

πάλιν means ‘again’; διὰ τὰ αὐτά, ‘for the same reasons’. The argument uses the same result three times. Each small diagonal lies on the whole diagonal when Κ lies on ΑΓ. Off that line, the small triangles no longer partition the two large halves in the required way.

Printed pages 100, 101.

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5. Add the equal small parts

ἐπεὶ οὖν τὸ μὲν ΑΕΚ τρίγωνον τῷ ΑΘΚ τριγώνῳ ἐστὶν ἴσον, τὸ δὲ ΚΖΓ τῷ ΚΗΓ, τὸ ΑΕΚ τρίγωνον μετὰ τοῦ ΚΗΓ ἴσον ἐστὶ τῷ ΑΘΚ τριγώνῳ μετὰ τοῦ ΚΖΓ·

iam quoniam △ AEK = AΘK et KZΓ = KHΓ, erit AEK + KHΓ = AΘK + KZΓ.

Since triangle AEK equals AΘK, and KZΓ equals KHΓ, triangle AEK together with KHΓ equals triangle AΘK together with KZΓ.

μέν … δέ pairs the two premises. μετά with the genitive joins areas. Notice the cross-pairing: one small triangle from each diagonal parallelogram is removed from each large half.

Printed pages 100, 101.

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6. Subtract, and keep the remainders

ἔστι δὲ καὶ ὅλον τὸ ΑΒΓ τρίγωνον ὅλῳ τῷ ΑΔΓ ἴσον· λοιπὸν ἄρα τὸ ΒΚ παραπλήρωμα λοιπῷ τῷ ΚΔ παραπληρώματί ἐστιν ἴσον.

uerum etiam ABΓ = AΔΓ. itaque etiam BK = KΔ.

The whole triangle ΑΒΓ also equals the whole triangle ΑΔΓ. Therefore the remaining complement ΒΚ equals the remaining complement ΚΔ.

ὅλον / ὅλῳ contrasts the wholes with λοιπὸν / λοιπῷ, the remainders. Common notion 3 justifies subtracting equal areas from equal totals. It does not require the removed shapes to be congruent to the remaining shapes.

Printed pages 100, 102, 101, 103.

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7. The general conclusion

Παντὸς ἄρα παραλληλογράμμου χωρίου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

Ergo in quouis parallelogrammo complementa parallelogrammorum circum diametrum positorum inter se aequalia sunt; quod erat demonstrandum.

Therefore, in every parallelogram, the complements of the parallelograms about the diagonal are equal to one another. This is what had to be shown.

δεῖξαι, ‘to show’, closes a theorem. The displayed numerical agreement illustrates the argument; the three diagonal bisections and subtraction explain it for every allowed parallelogram.

Printed pages 102, 103.

Complete II.5: Greek, Latin and English

1. One line, two cuts

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταξὺ τῶν τομῶν τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ἡμισείας τετραγώνῳ.

Si recta linea in partes aequales et inaequales secatur, rectangulum inaequalibus partibus totius comprehensum cum quadrato rectae inter sectiones positae aequale est quadrato dimidiae.

If a straight line is cut into equal and unequal parts, the rectangle contained by the unequal parts of the whole, together with the square on the distance between the cuts, equals the square on half the line.

τμηθῇ is an aorist passive subjunctive after ἐάν: ‘if it is cut’. There are two cuts of one line, not two different lines. μεταξύ / inter means ‘between’: the smaller square belongs to the distance between the cuts.

Printed pages 128, 129.

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2. Name the cuts and the claim

Εὐθεῖα γάρ τις ἡ ΑΒ τετμήσθω εἰς μὲν ἴσα κατὰ τὸ Γ, εἰς δὲ ἄνισα κατὰ τὸ Δ· λέγω, ὅτι τὸ ὑπὸ τῶν ΑΔ, ΔΒ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΓΔ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΒ τετραγώνῳ.

nam recta quaelibet AB in aequales partes secetur in Γ, in inaequales autem in Δ. dico, esse $\text{AΔ} \times \text{ΔB} + \text{ΓΔ}^2 = \text{ΓB}^2$.

Let a straight line AB be cut into equal parts at Γ and unequal parts at Δ. I say that the rectangle contained by AΔ and ΔB, together with the square on ΓΔ, equals the square on ΓB.

The displayed orientation follows the printed diagram: Δ lies between Γ and B. Reversing the chosen lengths turns the interpretation of the rectangle. The exactly equal case is a limiting identity; it does not satisfy this statement's unequal-cut premise.

Printed pages 128, 130, 129, 131.

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3. Build the square and its diagonal

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΓΒ τετράγωνον τὸ ΓΕΖΒ, καὶ ἐπεζεύχθω ἡ ΒΕ,

construatur enim in ΓB quadratum ΓEZB, et ducatur BE,

Draw the square ΓEZB on ΓB, and join BE.

ἀναγεγράφθω and ἐπεζεύχθω are perfect passive imperatives: let the figures stand drawn and joined. I.46 supplies construction of a square. Here BE is a diagonal of that square; the larger rectangle AΘ will appear outside part of it.

Printed pages 130, 131.

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4. Draw the parallels through the meeting point

καὶ διὰ μὲν τοῦ Δ ὁποτέρᾳ τῶν ΓΕ, ΒΖ παράλληλος ἤχθω ἡ ΔΗ, διὰ δὲ τοῦ Θ ὁποτέρᾳ τῶν ΑΒ, ΕΖ παράλληλος πάλιν ἤχθω ἡ ΚΜ, καὶ πάλιν διὰ τοῦ Α ὁποτέρᾳ τῶν ΓΛ, ΒΜ παράλληλος ἤχθω ἡ ΑΚ.

et per Δ utrique ΓΕ, ΒΖ parallela ducatur ΔΗ, per Θ autem utrique ΑΒ, ΕΖ parallela ducatur ΚΜ, et rursus per Α utrique ΓΛ, ΒΜ parallela ducatur ΑΚ.

Through Δ draw ΔΗ parallel to either ΓΕ or ΒΖ. Through Θ again draw ΚΜ parallel to either ΑΒ or ΕΖ, and through Α draw ΑΚ parallel to either ΓΛ or ΒΜ.

διά with the genitive means ‘through’. The diagram identifies Θ where ΔΗ meets ΒΕ. The parallels form the complements ΓΘ and ΘΖ. Greek ὁποτέρᾳ says ‘to either’; the printed Latin utrique says ‘to each/both’, which is consistent because the two named lines are parallel.

Printed pages 130, 131.

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5. The equal complements do the work

καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΓΘ παραπλήρωμα τῷ ΘΖ παραπληρώματι, κοινὸν προσκείσθω τὸ ΔΜ· ὅλον ἄρα τὸ ΓΜ ὅλῳ τῷ ΔΖ ἴσον ἐστίν.

et quoniam ΓΘ = ΘΖ, commune adiiciatur ΔΜ. itaque ΓΜ = ΔΖ.

Since the complement ΓΘ equals the complement ΘΖ, add the common region ΔΜ. The whole ΓΜ therefore equals the whole ΔΖ.

This is I.43, now used inside a square. κοινὸν προσκείσθω means ‘let the common part be added’. Both new totals contain ΔΜ. The lesson highlights the two equal complements first, then the larger equal totals.

Printed pages 130, 131.

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6. Use the equal halves of the original line

ἀλλὰ τὸ ΓΜ τῷ ΑΛ ἴσον ἐστίν, ἐπεὶ καὶ ἡ ΑΓ τῇ ΓΒ ἐστιν ἴση· καὶ τὸ ΑΛ ἄρα τῷ ΔΖ ἴσον ἐστίν.

uerum ΓΜ = ΑΛ, quoniam ΑΓ = ΓΒ. quare etiam ΑΛ = ΔΖ.

But ΓΜ equals ΑΛ, because ΑΓ equals ΓΒ. Therefore ΑΛ also equals ΔΖ.

The two horizontal rectangles have the same height and equal bases, since Γ bisects AB. Transitivity then replaces ΓΜ with ΑΛ in the preceding equality. The displayed outlines make each named region explicit.

Printed pages 130, 131.

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7. Add a complement to make a gnomon

κοινὸν προσκείσθω τὸ ΓΘ· ὅλον ἄρα τὸ ΑΘ τῷ ΜΝΞ γνώμονι ἴσον ἐστίν.

commune adiiciatur ΓΘ. itaque ΑΘ = ΜΝΞ gnomoni.

Add the common region ΓΘ. The whole ΑΘ therefore equals the gnomon ΜΝΞ.

γνώμονι is the dative of γνώμων; Latin gnomoni mirrors it. Here it names the L-shaped union ΓΘ + ΔΖ, not a sundial's upright. ΜΝΞ is the source's name for that region. Page 131 also uses M at the right boundary; the editor explicitly discusses the duplicated letter.

Printed pages 130, 131.

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8. Identify the rectangle's two sides

ἀλλὰ τὸ ΑΘ τὸ ὑπὸ τῶν ΑΔ, ΔΒ ἐστιν· ἴση γὰρ ἡ ΔΘ τῇ ΔΒ· καὶ ὁ ΜΝΞ ἄρα γνώμων ἴσος ἐστὶ τῷ ὑπὸ ΑΔ, ΔΒ.

uerum ΑΘ = ΑΔ × ΔΒ (nam ΔΘ = ΔΒ); quare etiam ΜΝΞ = ΑΔ × ΔΒ.

But ΑΘ is the rectangle contained by ΑΔ and ΔΒ, since ΔΘ equals ΔΒ. Therefore the gnomon ΜΝΞ also equals the rectangle contained by ΑΔ and ΔΒ.

The diagonal of the square makes the upper-right region ΔΜ a square, so its height ΔΘ equals its top side ΔΒ. Thus the outside rectangle really has the two unequal lengths from the opening statement.

Printed pages 130, 131.

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9. Fill the missing square

κοινὸν προσκείσθω τὸ ΛΗ, ὅ ἐστιν ἴσον τῷ ἀπὸ τῆς ΓΔ· ὁ ἄρα ΜΝΞ γνώμων καὶ τὸ ΛΗ ἴσα ἐστὶ τῷ ὑπὸ τῶν ΑΔ, ΔΒ περιεχομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς ΓΔ τετραγώνῳ.

commune adiiciatur ΛΗ, quod aequale est $\Gamma\Delta^2$. itaque ΜΝΞ + ΛΗ = ΑΔ × ΔΒ + $\Gamma\Delta^2$.

Add the common region ΛΗ, equal to the square on ΓΔ. The gnomon ΜΝΞ together with ΛΗ therefore equals the rectangle contained by ΑΔ and ΔΒ together with the square on ΓΔ.

The missing corner ΛΗ is a square. Its width is ΓΔ; its height is the same difference between the half-length and the short length. This square has area 6.25 in the opening 9 by 4 example.

Printed pages 130, 131.

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10. Recognize the completed whole

ἀλλὰ ὁ ΜΝΞ γνώμων καὶ τὸ ΛΗ ὅλον ἐστὶ τὸ ΓΕΖΒ τετράγωνον, ὅ ἐστιν ἀπὸ τῆς ΓΒ· τὸ ἄρα ὑπὸ τῶν ΑΔ, ΔΒ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΓΔ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΒ τετραγώνῳ.

sed ΜΝΞ + ΛΗ = ΓΕΖΒ = ΓΒ². itaque ΑΔ × ΔΒ + ΓΔ² = ΓΒ².

But the gnomon ΜΝΞ together with ΛΗ is the whole square ΓΕΖΒ, the square on ΓΒ. Therefore the rectangle contained by ΑΔ and ΔΒ, together with the square on ΓΔ, equals the square on ΓΒ.

The equal-area replacement becomes a statement about a complete square. For long side a and short side b, modern notation gives ab + ((a − b)/2)² = ((a + b)/2)². Euclid's argument works through figures and equal additions.

Printed pages 130, 131.

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11. Return to the general statement

Ἐὰν ἄρα εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταξὺ τῶν τομῶν τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ἡμισείας τετραγώνῳ· ὅπερ ἔδει δεῖξαι.

Ergo si recta linea in partes aequales et inaequales secatur, rectangulum partibus inaequalibus totius comprehensum cum quadrato rectae inter sectiones positae aequale est quadrato dimidiae; quod erat demonstrandum.

Therefore, if a straight line is cut into equal and unequal parts, the rectangle contained by the unequal parts of the whole, together with the square on the distance between the cuts, equals the square on half the line. This is what had to be shown.

The conclusion repeats the opening condition. II.14 later applies this result to the extended line of its semicircle construction. Its different point names do not change the relation.

Printed pages 132, 133.

One letter, two uses

Printed page 131 labels the boundary point M and also names the gnomon MNΞ. In his footnote Heiberg says that M is used twice in the diagram received from the Basel edition; he records Gregory's NΞO alternative but declines to change the reading against the manuscripts. This workshop retains M at the boundary and puts the region name MNΞ in the explanation. It does not invent a single point that resolves both uses.

Source and model boundaries

J. L. Heiberg, *Euclidis opera omnia*, volume I, Leipzig: Teubner, 1883, pp. 100–103 and 128–133. University of Toronto scan, public domain. Greek ϑ is normalized to θ; line breaks, critical apparatus and inline Latin references are omitted in the reading. The source images preserve them. Latin is Heiberg's nineteenth-century translation. New English, divisions, notes, coordinates, colors and off-diagonal experiment are modern teaching work.

I.34, I.46 and earlier parallel constructions are cited prerequisites, not complete proofs reproduced here. The square is a special parallelogram; the I.43 experiment also permits oblique ones. In II.5 the longer segment is placed first, as in the diagram. Equal input lengths are explicitly a limiting case. The diagrams use equal horizontal and vertical scales; values are rounded for display. The numerical checks illustrate, rather than replace, the source's proof.

This file embeds every source image and its font and makes no remote subresource requests. Opening the application link needs a connection.

ληλόγραμμον συνέσταται τὸ ΖΕΓΗ ἐν γωνίᾳ τῇ ὑπὸ ΓΕΖ, ἥτις ἐστὶν ἴση τῇ Δ· ὅπερ ἔδει ποιῆσαι.

μγ΄.

Παντὸς παραλληλογράμμου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν.

Ἔστω παραλληλόγραμμον τὸ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ ἡ ΑΓ, περὶ δὲ τὴν ΑΓ παραλληλόγραμμα μὲν ἔστω τὰ ΕΘ, ΖΗ, τὰ δὲ λεγόμενα παραπληρώματα τὰ ΒΚ, ΚΔ· λέγω, ὅτι ἴσον ἐστὶ τὸ ΒΚ παραπλήρωμα τῷ ΚΔ παραπληρώματι.

Ἐπεὶ γὰρ παραλληλόγραμμόν ἐστι τὸ ΑΒΓΔ, διάμετρος δὲ αὐτοῦ ἡ ΑΓ, ἴσον ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΓΔ τριγώνῳ. πάλιν, ἐπεὶ παραλληλόγραμμόν ἐστι τὸ ΕΘ, διάμετρος δὲ αὐτοῦ ἐστιν ἡ ΑΚ, ἴσον ἐστὶ τὸ ΑΕΚ τρίγωνον τῷ ΑΘΚ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΚΖΓ τρίγωνον τῷ ΚΗΓ ἐστιν ἴσον. ἐπεὶ οὖν τὸ μὲν ΑΕΚ τρίγωνον τῷ ΑΘΚ τριγώνῳ ἐστὶν ἴσον, τὸ δὲ ΚΖΓ τῷ ΚΗΓ, τὸ ΑΕΚ τρίγωνον μετὰ τοῦ ΚΗΓ ἴσον ἐστὶ τῷ ΑΘΚ τριγώνῳ μετὰ τοῦ ΚΖΓ· ἔστι δὲ καὶ ὅλον τὸ ΑΒΓ τρίγωνον ὅλῳ τῷ ΑΔΓ ἴσον· λοιπὸν ἄρα τὸ

XLIII. Boetius p. 384, 10. Apud Proclum excidit.

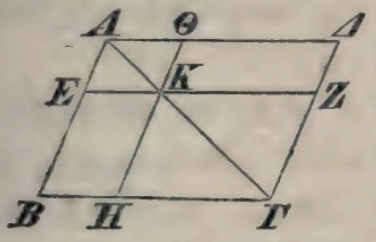
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1. *συνέσταται*] PBFbp; *συνίσταται* V; *συνεστάθη* φ. ΖΕΓΗ] e corr. φ. *ἐν γωνίᾳ τῇ ὑπὸ ΓΕΖ*] om. F (mg. m. rec. *ἐν γωνίᾳ τῇ ὑπὸ* ΖΕΓ *ἥ ἐστιν*). 2. ΓΕΖ] seq. ras. 1 litt. P; ΖΕΓ B, V m. 2. *ἥτις*] PVp; *ἥ* BFb. *ποιῆσαι*] in ras. p; *δεῖξαι* P (*ἐν ἄλλῳ δεῖξαι* mg. b). 3. *διάμετρον αὐτοῦ* p. 8. Post *τὴν ΑΓ* in V m. 2 add. *διάμετρον*. 9. ΖΗ] ΗΖ F. *παραπληρώματα*] *-πληρώματα* in ras. m. 2 V. *τά*] m. rec. P. 10. *ἐστίν* P. 11. *παραπληρώματι*] *παρα-* supra V m. 2. 13. *ἡ*] *ἐστιν ἡ* F. *ἴσον*] *ἴσον ἄρα* F.

mum constructum est $ZE\Gamma H$ in angulo ΓEZ, qui aequalis est angulo Δ; quod oportebat fieri.

XLIII.

In quouis parallelogrammo complementa parallelogrammorum circum diametrum positorum inter se aequalia sunt.



Sit parallelogrammum $AB\Gamma\Delta$, diametrus autem eius $A\Gamma$, et circum $A\Gamma$ parallelogramma sint $E\Theta$, ZH, et complementa, quae uocantur, BK, $K\Delta$. dico, esse

$$BK = K\Delta.$$

nam quoniam parallelogrammum est $AB\Gamma\Delta$, diametrus autem eius $A\Gamma$, erit $\triangle AB\Gamma = A\Gamma\Delta$ [prop. XXXIV]. rursus quoniam parallelogrammum est $E\Theta$, diametrus autem eius AK, erit $\triangle AEK = A\Theta K$. eadem de causa etiam $KZ\Gamma = KH\Gamma$ [id.]. iam quoniam $\triangle AEK = A\Theta K$ et $KZ\Gamma = KH\Gamma$, erit

$$AEK + KH\Gamma = A\Theta K + KZ\Gamma \text{ [}\kappa. \ \ddot{\epsilon}\nu\nu. \text{ 2]}.$$

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14. *ἐστιν* P. 15. *EΘ*] P m. 1, Bp, V m. 2; *AKEΘ* P m. rec.; *AEKΘ* F (*AEK* in ras.), V m. 1, b, Zambertus. *ἐστιν*] PFB; om. Vbp. *ἴσον ἄρα ἐστίν* P. 16. *AEK*] *AΓE* F; corr. in *AKE* m. 2. *AΘK*] *ΘK* litt. in ras. V. *τὰ αὐτά*] *ταῦτα* BVb. 17. *KZΓ*] *KHΓ* p. *KHΓ*] *KΓZ* p. Dein add. *τριγώνῳ* P m. 2, FVbp. *ἴσον ἐστίν* Vb. 18. *AEK*] *E* litt. e corr. F. *τρίγωνον*] supra m. 2 V. *AΘK*] litt. *ΘK* in ras. V. *τριγώνῳ*] om. p. 19. *ἴσον ἐστί* Vb. *KZΓ*] *KHΓ* p. *KHΓ*] litt. *H* eras. F; *KΓZ* p. Post *τό* add. b *ἄρα* comp. m. 1. *AEK*] *E* litt. in ras. F. *τὸ AEK* — 21. *KZΓ*] mg. m. 1 P. 20. *τρίγωνον*] comp. supra m. 2 V. *KHΓ*] corr. ex *KEΓ* m. 2 F. *ἐστίν* Fp. *ἐστιν ἴσον* b. 22. *AΔΓ*] litt. *Δ* e corr. F.

ΒΚ παραπλήρωμα λοιπῷ τῷ ΚΔ παραπληρώματί ἐστιν ἴσον.

Παντὸς ἄρα παραλληλογράμμου χωρίου τῶν περὶ τὴν διάμετρον παραλληλογράμμων τὰ παραπληρώματα ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

μδ'.

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι τριγώνῳ ἴσον παραλληλόγραμμον παραβαλεῖν ἐν τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ δοθὲν τρίγωνον τὸ Γ, ἡ δὲ δοθεῖσα γωνία εὐθύγραμμος ἡ Δ· δεῖ δὴ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν ΑΒ τῷ δοθέντι τριγώνῳ τῷ Γ ἴσον παραλληλόγραμμον παραβαλεῖν ἐν ἴσῃ τῇ Δ γωνίᾳ.

Συνεστάτω τῷ Γ τριγώνῳ ἴσον παραλληλόγραμμον τὸ ΒΕΖΗ ἐν γωνίᾳ τῇ ὑπὸ ΕΒΗ, ἣ ἐστιν ἴση τῇ Δ· καὶ κείσθω ὥστε ἐπ' εὐθείας εἶναι τὴν ΒΕ τῇ ΑΒ, καὶ διήχθω ἡ ΖΗ ἐπὶ τὸ Θ, καὶ διὰ τοῦ Α ὁποτέρᾳ τῶν ΒΗ. ΕΖ παράλληλος ἤχθω ἡ ΑΘ. καὶ ἐπεζεύχθω ἡ ΘΒ. καὶ ἐπεὶ εἰς παραλλήλους τὰς ΑΘ. ΕΖ εὐθεῖα ἐνέπεσεν ἡ ΘΖ. αἱ ἄρα ὑπὸ ΑΘΖ, ΘΖΕ γωνίαι δυσὶν ὀρθαῖς εἰσιν ἴσαι. αἱ ἄρα ὑπὸ ΒΘΗ, ΗΖΕ δύο ὀρθῶν ἐλάσσονές εἰσιν· αἱ δὲ ἀπὸ ἐλασσόνων ἢ δύο ὀρθῶν εἰς ἄπειρον ἐκβαλλόμεναι συμπίπτουσιν·

XLIV. Boetius p. 384, 14.

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1. *ἴσον ἐστίν* p. 3. *χωρίον*] om. BVp; cfr. p. 100, 4. *διάμετρον αὐτοῦ* p. 8. *παραβαλεῖν*] *-βαλ-* in ras. m. 1 B. *ἐν*] *ἐν γωνίᾳ, ἣ ἐστιν ἴσῃ* Proclus: cfr. Campanus. 12. *εὐθεῖαν*] mg. m. 1 F. 17. *ὥστ'* V. 18. *ΑΒ*] *ΑΘ* π. 19. *ΒΗ*] seq. ras. 1 litt. F. *ΑΘ*] *ΑΒ* F. *καί* — 20. *ΘΒ*] mg. m. 1 P. 20. *ΘΒ*] *ΒΘ* F. 21. *εὐθείας* BVp. *ἐν-*

uerum etiam $AB\Gamma = A\Delta\Gamma$. itaque etiam

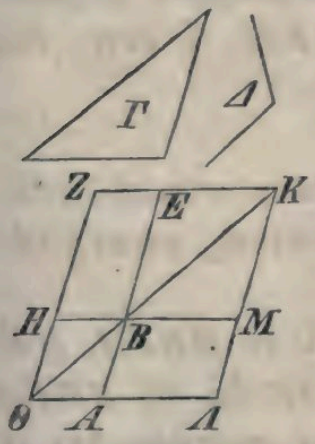
$$BK = K\Delta \text{ [}\varkappa. \ \check{\varepsilon}\nu\nu. \ 3\text{]}.$$

Ergo in quouis parallelogrammo complementa parallelogrammorum circum diametrum positorum inter se aequalia sunt; quod erat demonstrandum.

XLIV.

Datae rectae parallelogrammum dato triangulo aequale adplicare in dato angulo rectilineo.

Sit data recta AB, datus autem triangulus Γ, datus autem angulus rectilineus Δ. oportet igitur datae rectae AB parallelogrammum dato triangulo Γ aequale adplicare in angulo aequali angulo Δ.



construatur parallelogrammum $BEZH$ triangulo Γ aequale in angulo EBH, qui aequalis est angulo Δ [prop. XLII], et ponatur ita, ut BE, AB in eadem recta sint, et educatur ZH ad Θ, et per A utrique BH, EZ parallela ducatur $A\Theta$ [prop. XXXI], et ducatur ΘB. et quoniam in parallelas $A\Theta$, EZ recta incidit ΘZ,

$$\angle A\Theta Z + \Theta ZE$$

duobus rectis aequales erunt [prop. XXIX]. itaque

$$\angle B\Theta H + HZE$$

duobus rectis minores erunt; quae autem ex angulis minoribus, quam sunt duo recti, in infinitum producuntur,

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ἔπεσεν] P; *ἐμπέπτωκεν* Theon (BFVbp); cfr. p. 106, 14. 108, 25. *ἄρα*] om. P. $A\Theta Z$] $BH\Theta$ p; corr. m. rec. ΘZE — 22. $B\Theta H$] mg. m. rec. p. 22. *εἰσιν ἴσαι*] PBF; *ἴσαι εἰσίν* Vbp. Ante *αἱ* insert. comp. *καί* B. $B\Theta Z$, ΘZE P. 23. *ἀπό*] *ἀπ'* p. 24. *ἐκβαλλόμεναι εἰς ἄπειρον* p. *ἐκβαλόμεναι* P.

ἔστι δὲ καὶ τὰ ΘΖ, ΓΚ τετράγωνα ἀπὸ τῶν ΑΓ, ΓΒ· τὰ ἄρα τέσσαρα τὰ ΘΖ, ΓΚ, ΑΗ, ΗΕ ἴσα ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ δὶς ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογωνίῳ. ἀλλὰ τὰ ΘΖ, ΓΚ, ΑΗ, ΗΕ ὅλον ἐστὶ τὸ ΑΔΕΒ, ὅ ἐστιν ἀπὸ τῆς ΑΒ τετράγωνον· τὸ ἄρα ἀπὸ τῆς ΑΒ τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν ΑΓ, ΓΒ τετραγώνοις καὶ τῷ δὶς ὑπὸ τῶν ΑΓ, ΓΒ περιεχομένῳ ὀρθογωνίῳ.

Ἐὰν ἄρα εὐθεῖα γραμμὴ τμηθῇ, ὡς ἔτυχεν, τὸ ἀπὸ τῆς ὅλης τετράγωνον ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν τμημάτων τετραγώνοις καὶ τῷ δὶς ὑπὸ τῶν τμημάτων περιεχομένῳ ὀρθογωνίῳ· ὅπερ ἔδει δεῖξαι.

[Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἐν τοῖς τετραγώνοις χωρίοις τὰ περὶ τὴν διάμετρον παραλληλόγραμμα τετράγωνά ἐστιν].

ε'.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταξὺ τῶν τομῶν τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ἡμισείας τετραγώνῳ.

Εὐθεῖα γάρ τις ἡ ΑΒ τετμήσθω εἰς μὲν ἴσα κατὰ

IV. πόρ. De Proclo p. 304 u. ad IV, 15. V. Boetius p. 385, 17.

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1. ἔστιν P. τά] τό F; corr. m. 2. τετράγωνον F; corr. m. 2. 2. τά] (alt.) om. F. ἐστίν P. 3. τε] m. 2 V. 4. ὀρθογώνια φ. τά] τὰ τέσσαρα P. ΘΖ] Θ in ras. V; ΖΘ B. 5. ΗΕ] Η e corr. p. ἐστίν P. ΑΔΕΒ

(nam $H\Gamma = \Gamma B$), erit etiam $HE = A\Gamma \times \Gamma B$. itaque $AH + HE = 2\, A\Gamma \times \Gamma B$. uerum etiam quadrata ΘZ, ΓK in $A\Gamma$, ΓB constructa sunt. ergo $\Theta Z + \Gamma K + AH + HE = A\Gamma^2 + \Gamma B^2 + 2\, A\Gamma \times \Gamma B$. sed $\Theta Z + \Gamma K + AH + HE = A\Delta EB = AB^2$. itaque $AB^2 = A\Gamma^2 + \Gamma B^2 + 2\, A\Gamma \times \Gamma B$.

Ergo si recta linea utcunque secatur, quadratum totius aequale est quadratis partium et duplo rectangulo partibus comprehenso; quod erat demonstrandum.[1])

V.

Si recta linea in partes aequales et inaequales secatur, rectangulum inaequalibus partibus totius comprehensum cum quadrato rectae inter sectiones positae aequale est quadrato dimidiae.[2])

nam recta quaelibet AB in aequales partes sece-

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1) Etiam Campanus hic duas demonstrationes habet, quarum prior reiectae, altera neque huic neque reiectae similis est. de hac habet: „sed hac uia non patet correlarium, sicut uia praecedenti patet, unde prima est autori magis consona.“ nam corollarium et ipse habet. itaque fortasse Theone antiquius est.

2) $ab + \left(\frac{a+b}{2} - b\right)^2 = \left(\frac{a+b}{2}\right)^2$.

τετράγωνον V. 6. *AB τετράγωνον*] (prius) mg. m. 2 V; in textu ras. 2—3 litt. *τετράγωνον*] mg. m. 2 F. 7. *ἐστίν* P. *τε*] om. p. *τῶν*] m. 2 F. 9. *ἔτυχεν*] B; *ἔτυχε* uulgo. 10. *ἐστίν* P. *τε*] om. p. 12. Sequitur alia demonstratio, quam Augustum secutus in appendicem reieci. 13. *πόρισμα* — 16. *ἐστιν*] add. Theon? (BFVp); mg. m. rec. P. 14. *τούτων* P. *φανερόν ἐστιν* V. 18. *εἰς*] supra m. 1 V. 19. *εἰς ἄνισα* p. 21. *ἐστίν* P.

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τὸ Γ, εἰς δὲ ἄνισα κατὰ τὸ Δ· λέγω, ὅτι τὸ ὑπὸ τῶν ΑΔ, ΔΒ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΓΔ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΒ τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΓΒ τετράγωνον τὸ ΓΕΖΒ, καὶ ἐπεζεύχθω ἡ ΒΕ, καὶ διὰ μὲν τοῦ Δ ὁποτέρᾳ τῶν ΓΕ, ΒΖ παράλληλος ἤχθω ἡ ΔΗ, διὰ δὲ τοῦ Θ ὁποτέρᾳ τῶν ΑΒ, ΕΖ παράλληλος πάλιν ἤχθω ἡ ΚΜ, καὶ πάλιν διὰ τοῦ Α ὁποτέρᾳ τῶν ΓΔ, ΒΜ παράλληλος ἤχθω ἡ ΑΚ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΓΘ παραπλήρωμα τῷ ΘΖ παραπληρώματι, κοινὸν προσκείσθω τὸ ΔΜ· ὅλον ἄρα τὸ ΓΜ ὅλῳ τῷ ΔΖ ἴσον ἐστίν. ἀλλὰ τὸ ΓΜ τῷ ΑΛ ἴσον ἐστίν, ἐπεὶ καὶ ἡ ΑΓ τῇ ΓΒ ἐστιν ἴση· καὶ τὸ ΑΛ ἄρα τῷ ΔΖ ἴσον ἐστίν. κοινὸν προσκείσθω τὸ ΓΘ· ὅλον ἄρα τὸ ΑΘ τῷ ΜΝΞ γνώμονι ἴσον ἐστίν. ἀλλὰ τὸ ΑΘ τὸ ὑπὸ τῶν ΑΔ, ΔΒ ἐστιν· ἴση γὰρ ἡ ΔΘ τῇ ΔΒ· καὶ ὁ ΜΝΞ ἄρα γνώμων ἴσος ἐστὶ τῷ ὑπὸ ΑΔ, ΔΒ. κοινὸν προσκείσθω τὸ ΛΗ, ὅ ἐστιν ἴσον τῷ ἀπὸ τῆς ΓΔ· ὁ ἄρα ΜΝΞ γνώμων καὶ τὸ ΛΗ ἴσα ἐστὶ τῷ ὑπὸ τῶν ΑΔ, ΔΒ περιεχομένῳ ὀρθογωνίῳ καὶ τῷ ἀπὸ τῆς ΓΔ τετραγώνῳ. ἀλλὰ ὁ ΜΝΞ γνώμων καὶ τὸ ΛΗ ὅλον ἐστὶ τὸ ΓΕΖΒ τετράγωνον, ὅ ἐστιν ἀπὸ τῆς ΓΒ· τὸ ἄρα ὑπὸ τῶν ΑΔ, ΔΒ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΓΔ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΒ τετραγώνῳ.

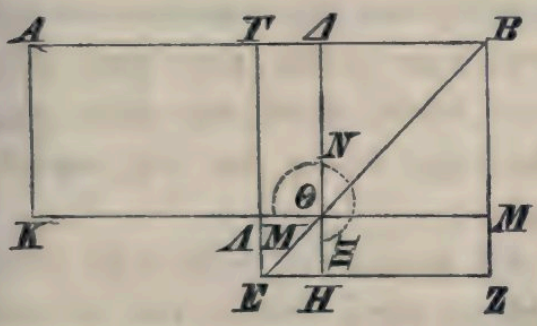
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3. ἐστίν P. τετραγώνῳ] om. B; comp. add. m. 2 F. 5. ΓΕΖΒ] in ras. p. ΒΕ] Β in ras. F. 6. ΒΖ] ΖΒ F. διὰ δέ] καὶ διά V. 7. πάλιν] om. p, m. 2 V. 8. καὶ πάλιν — 9. ἡ ΑΚ] mg. m. rec. P. 10. ΘΖ] ΖΘ F. 12. ἴσον ἐστίν] (alt.) ἐστιν ἴσον V. 13. ἐπεί — ἴση] mg. m. 2 V (ἴση ἐστί). 14. ἐστιν ἴσον V. ἐστίν] P, comp. m. 2 F; ἐστί Bp. 15.

tur in I, in inaequales autem in Δ. dico, esse

$$A\Delta \times \Delta B + \Gamma\Delta^2 = \Gamma B^2.$$

construatur enim in ΓB quadratum ΓEZB [I, 46], et ducatur BE, et per Δ utrique ΓE, BZ parallela ducatur ΔH, per Θ autem utrique AB, EZ parallela ducatur KM [I, 30. 31], et rursus per A utrique $\Gamma\Delta$, BM parallela ducatur AK. et quoniam $\Gamma\Theta = \Theta Z$ [I, 43], commune adiiciatur ΔM. itaque $\Gamma M = \Delta Z$. uerum



$\Gamma M = A\Lambda$, quoniam $A\Gamma = \Gamma B$. quare etiam $\Lambda\Lambda = \Delta Z$. commune adiiciatur $\Gamma\Theta$. itaque $A\Theta = MN\Xi$ gnomoni.[1]) uerum

$$A\Theta = A\Delta \times \Delta B$$

(nam $\Delta\Theta = \Delta B$); quare etiam $MN\Xi = A\Delta \times \Delta B$. commune adiiciatur ΛH, quod aequale est $\Gamma\Delta^2$. itaque $MN\Xi + \Lambda H = A\Delta \times \Delta B + \Gamma\Delta^2$. sed

$$MN\Xi + \Lambda H = \Gamma EZB = \Gamma B^2.$$

itaque $A\Delta \times \Delta B + \Gamma\Delta^2 = \Gamma B^2$.

1) Cum littera M in figura, quam ex ed. Basil. recepimus, bis usurpetur, non sine causa pro $MN\Xi$ a Gregorio scriptum est $N\Xi O$, ut prop. VI. sed non audeo contra codd. mutare.

$MN\Xi$ *γνώμονι*] P; Campanus; ΔZ *καὶ* $\Delta\Lambda$ Theon (BFV; pro $\Delta\Lambda$ in F ΔA; $\Delta\Lambda$ *καὶ* ΔZ p). *τὸ* $A\Theta$] *τὸ μὲν* $A\Theta$ Bp. 16. *γὰρ ἡ*] *ἡ γάρ* P. $\Delta\Theta$] ΔB p. ΔB] $\Delta\Theta$ *ἐστι* p. Post ΔB add. Theon: *τὰ δὲ* $Z\Delta$, $\Delta\Lambda$ *ἐστιν ὁ* $MN\Xi$ *γνώμων* B ($Z\Delta\Lambda$), F, V (prius Δ in ras.), p (*ὁ* $MN\Xi$ *ἐστι*); om. P. 17. *καί*] om. p. *τῷ*] *τό* F. *ὑπὸ τῶν* p. 19. *ἐστίν* P. 20. *περιεχομένων ὀρθογωνίων* F. 21. *ἀλλὰ*] *ἀλλ'* F; *ἀλλὰ καί* V. 23. ΓB] post ras. 1 litt. V; $B\Gamma$ p. 24. *ἀπὸ τῆς*] supra m. 2 F; *ἀπό* P. *ἐστίν* PV.

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Ἐὰν ἄρα εὐθεῖα γραμμὴ τμηθῇ εἰς ἴσα καὶ ἄνισα, τὸ ὑπὸ τῶν ἀνίσων τῆς ὅλης τμημάτων περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς μεταξὺ τῶν τομῶν τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ἡμισείας τετραγώνῳ. ὅπερ ἔδει δεῖξαι.

ε'.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ δίχα, προστεθῇ δέ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας, τὸ ὑπὸ τῆς ὅλης σὺν τῇ προσκειμένῃ καὶ τῆς προσκειμένης περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς συγκειμένης ἔκ τε τῆς ἡμισείας καὶ τῆς προσκειμένης τετραγώνῳ.

Εὐθεῖα γάρ τις ἡ AB τετμήσθω δίχα κατὰ τὸ $Γ$ σημεῖον, προσκείσθω δέ τις αὐτῇ εὐθεῖα ἐπ' εὐθείας ἡ $BΔ$· λέγω, ὅτι τὸ ὑπὸ τῶν $AΔ$, $ΔB$ περιεχόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς $ΓB$ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς $ΓΔ$ τετραγώνῳ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς $ΓΔ$ τετράγωνον τὸ $ΓEZΔ$, καὶ ἐπεζεύχθω ἡ $ΔE$, καὶ διὰ μὲν τοῦ B σημείου ὁποτέρᾳ τῶν $EΓ$, $ΔZ$ παράλληλος ἤχθω ἡ BH, διὰ δὲ τοῦ $Θ$ σημείου ὁποτέρᾳ τῶν AB, EZ παράλληλος ἤχθω ἡ KM, καὶ ἔτι διὰ τοῦ A ὁποτέρᾳ τῶν $ΓΔ$, $ΔM$ παράλληλος ἤχθω ἡ AK.

Ἐπεὶ οὖν ἴση ἐστὶν ἡ $AΓ$ τῇ $ΓB$, ἴσον ἐστὶ καὶ τὸ $AΛ$ τῷ $ΓΘ$. ἀλλὰ τὸ $ΓΘ$ τῷ $ΘZ$ ἴσον ἐστίν. καὶ

VI. Schol. in Archim. III p. 383. Boetius p. 385, 22.

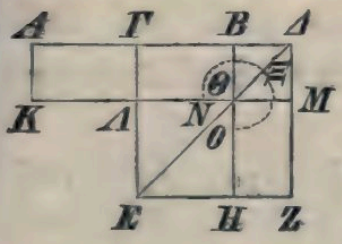
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1. γραμή P. εἰς ἄνισα p. 4. ἐστίν PV. 8. ἐπ' εὐθείας, τὸ ὑπό] in ras. V. 9. προσκειμένῃ] -σ- supra p. προκειμένης V, et p sed corr. m. 1. 11. ἐστίν V. 12. προσκειμένης] -σ- insert. p. Post hoc uerbum legitur ὡς ἀπὸ

Ergo si recta linea in partes aequales et inaequales secatur, rectangulum partibus inaequalibus totius comprehensum cum quadrato rectae inter sectiones positae aequale est quadrato dimidiae; quod erat demonstrandum.

VI.

Si recta linea in duas partes aequales secatur, et alia quaedam recta ei in directum adiicitur, rectangulum tota cum adiecta et adiecta comprehensum cum quadrato dimidiae aequale est quadrato in dimidia adiectaque descripto.[1])



nam recta aliqua AB in duas partes aequales secetur in puncto Γ, et alia quaedam recta $B\Delta$ ei in directum adiiciatur. dico, esse

$$A\Delta \times \Delta B + \Gamma B^2 = \Gamma\Delta^2.$$

construatur enim in $\Gamma\Delta$ quadratum $\Gamma EZ\Delta$, et ducatur ΔE, et per B punctum utrique $E\Gamma$, ΔZ parallela ducatur BH, per Θ autem punctum utrique AB, EZ parallela ducatur KM, et praeterea per A utrique $\Gamma\Lambda$, ΔM parallela ducatur AK. iam quoniam $A\Gamma = \Gamma B$, erit etiam $A\Lambda = \Gamma\Theta$. sed $\Gamma\Theta = \Theta Z$ [I, 43]. quare etiam $A\Lambda = \Theta Z$. commune adiiciatur ΓM.

1) $(2a + b)\,b + a^2 = (a + b)^2$.

μιᾶς ἀναγραφέντι in p, P mg. m. rec., Zamberto; om. Boetius, Campanus, P m. 1, B, V m. 1; in F fuit a m. 1 (restant . . αγραφέντι), sed τετραγώνῳ φ; ὡς ἀπὸ μιᾶς V mg. m. 2. 18. ἐστίν V. 20. ἐπεζευχθω — 21. ΔZ] mg. m. rec. P. 21. EΓ] ΓE Pp. ΔZ] ZΔ φ. 22. σημείου] om. p. AB] ABΔ p, AΔ P. 25. AΓ] in ras. V. ἐστίν V. 26. ἀλλά] ἀλλὰ καί F. ἴσον ἐστίν] P; ἴσον F, ἴσον ἐστί B; ἐστιν ἴσον Vp.

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